Cambridge Assessment International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*7890471656*

**MATHEMATICS (SYLLABUS D)** **4024/12**

Paper 1 **October/November 2020**

**2 hours**

You must answer on the question paper.

You will need: Geometrical instruments

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

**INFORMATION**

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Blank pages are indicated.

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**ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER**

**1** **(a)** Evaluate $\frac{4}{5} - \frac{2}{3}$.

..... [1]

**(b)** Evaluate $2.7 \times 0.2$ .

..... [1]

**2** Find the fraction which lies exactly halfway between $\frac{3}{5}$ and $\frac{5}{7}$.
Give your answer in its simplest form.

..... [2]

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**3** Factorise.

**(a)** $12t^2 - 4t$

..... [1]

**(b)** $a(x-y) + b(y-x)$

..... [1]

**(c)** $x^2 - 2x - 3$

..... [1]

**4** Write these lengths in order of size, starting with the smallest.

0.043 km  433 cm  4340 mm  $4\frac{1}{3}$ m

..... , ..... , ..... , ..... [2]
*smallest*



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**5** Sandra buys a vase for $40 and sells it for $200.

Calculate her percentage profit.

..... % [2]

**6** These are the minimum temperatures, in °C, recorded by a weather station each day during one week.

−2.3 −4.6 −1.2 −0.7 −1.4 −2.4 −3.5

**(a)** Find the range of these temperatures.

..... °C [1]

**(b)** How many of these temperatures are between −4°C and −2°C?

..... [1]

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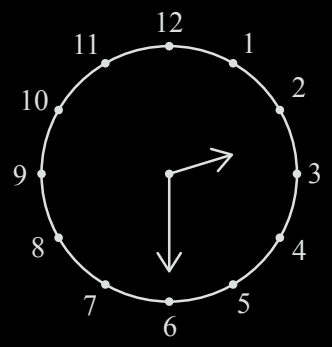

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**7** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{6.044^2}{212 \times 0.304}.$$

..... [2]

**8**




NOT TO SCALE

In the diagram, the time on the clock is 2.30 pm.

Calculate the **reflex** angle between the two hands of the clock.

..... [2]



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**9** **(a)** Simplify $3(3a-4)+2(2-a)$.

..... [2]

**(b)** Given that $4x = 3y$, find the numerical value of $\frac{8x+y}{y}$.

..... [1]

**10** Solve the simultaneous equations.

$$3x-2y = 12$$
$$4x+\ y = 5$$

$x =$ .....

$y =$ ..... [3]

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**11** **(a)** Express $340\,000$ in standard form.

..... [1]

**(b)** Evaluate $\dfrac{4\times10^{7}}{8\times10^{21}}$, giving your answer in standard form.

..... [2]

**(c)** $7\times10^{a}-3\times10^{a-1}=k\times10^{a}$

Find $k$.

$k=$ ..... [1]

**12** **(a)** Simplify $(2x^{2})^{3}$.

..... [1]

**(b)** Simplify $6t^{3}\div\left(\dfrac{2}{3}t^{2}\right)$.

..... [2]

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**13** **(a)** $P = \{ 1, 2, 3, 4, 5, 6, 7, 8 \}$

$Q = \{ 1, 3, 5, 7, 9, 11 \}$

Find $\mathrm{n}(P \cup Q)$.

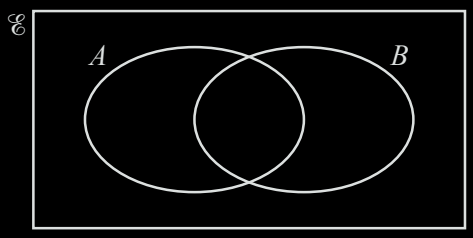
..... [1]

**(b)** $p \in A \cap B$

$q \in (A \cup B)'$

$r \in A \cap B'$

On the Venn diagram below, write each of the letters $p$, $q$ and $r$ in its appropriate subset.



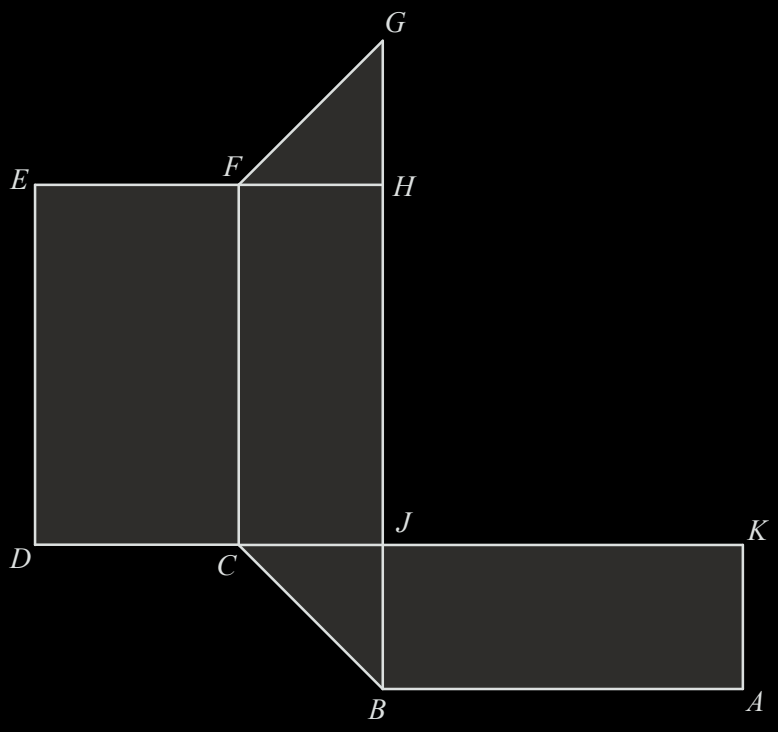

[3]

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**14**




This net is folded to make a triangular prism.

**(a)** Which vertices join with $A$?

..... [1]

**(b)** Which edge joins with $DE$?

..... [1]

**(c)** $FH = 2$ cm, $GH = 2$ cm and $JH = 5$ cm.

Find the volume of the triangular prism.

..... $cm^3$ [2]

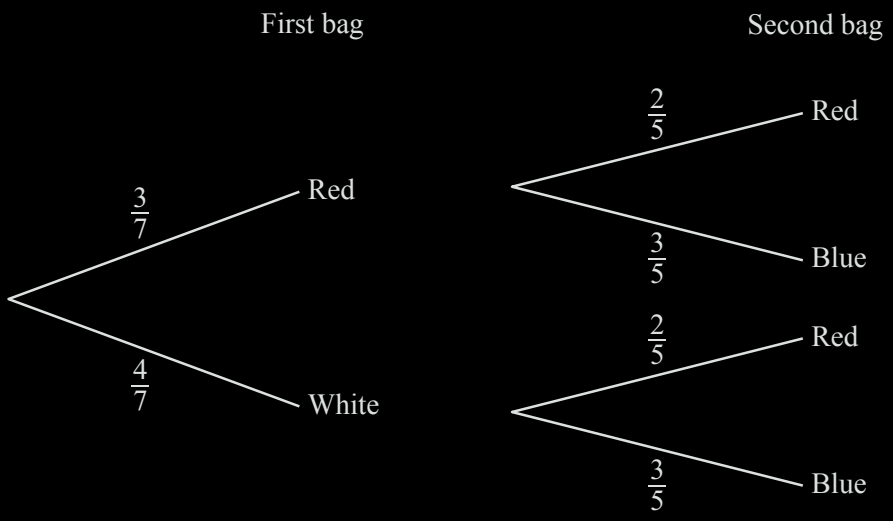


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**15** Two bags contain beads.

The first bag contains 7 beads, of which 3 are red and 4 are white.
The second bag contains 5 beads, of which 2 are red and 3 are blue.

One bead is taken, at random, from each bag.
The tree diagram is shown below.




Find the probability that

**(a)** both beads are red,

..... [1]

**(b)** both beads are white,

..... [1]

**(c)** **exactly** one bead is red.

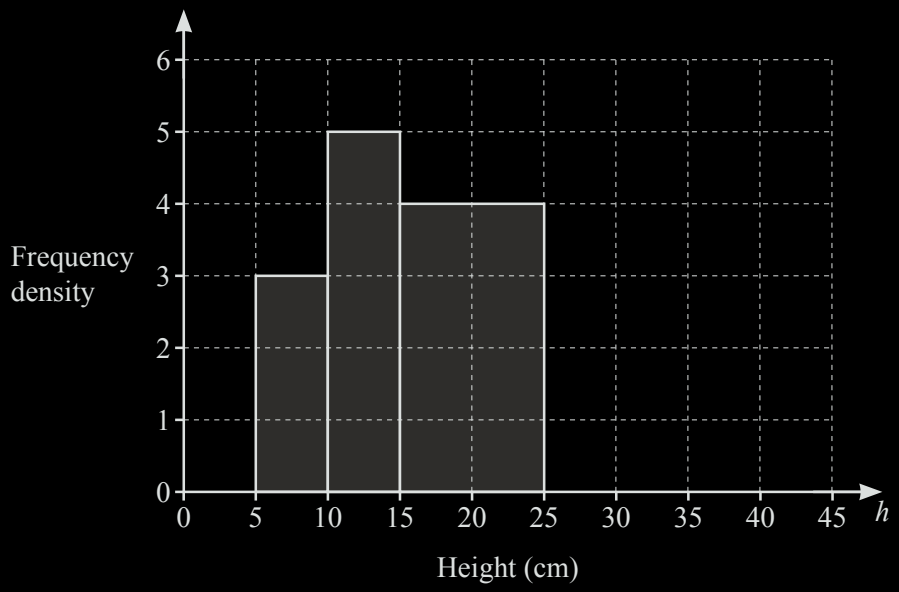
..... [2]

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**16** The heights of a sample of plants were measured.
The results are shown in the table and in the histogram.

| Height ($h$ cm) | $5 < h \leqslant 10$ | $10 < h \leqslant 15$ | $15 < h \leqslant 25$ | $25 < h \leqslant 40$ |
|---|---|---|---|---|
| Frequency | 15 | 25 | $p$ | 30 |




**(a)** Use the histogram to find the value of $p$.

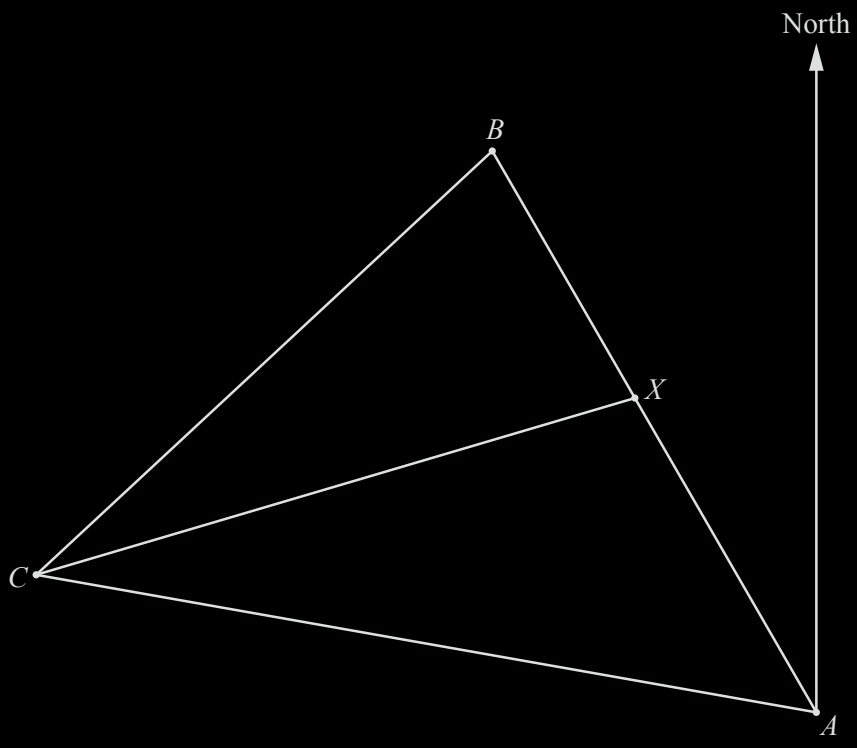
$p =$ ..... [1]

**(b)** Complete the histogram. [1]



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**17** The diagram shows the positions of three boats $A$, $B$ and $C$.




**(a)** By measurement, find the bearing of $B$ from $A$.

..... [1]

**(b)** $CX$ is the bisector of angle $ACB$.

**(i)** **Using compasses and a straight edge only**, construct the locus of points inside triangle $ABC$ that are equidistant from $B$ and $C$. [2]

**(ii)** A ship is

- nearer to $AC$ than to $BC$

and

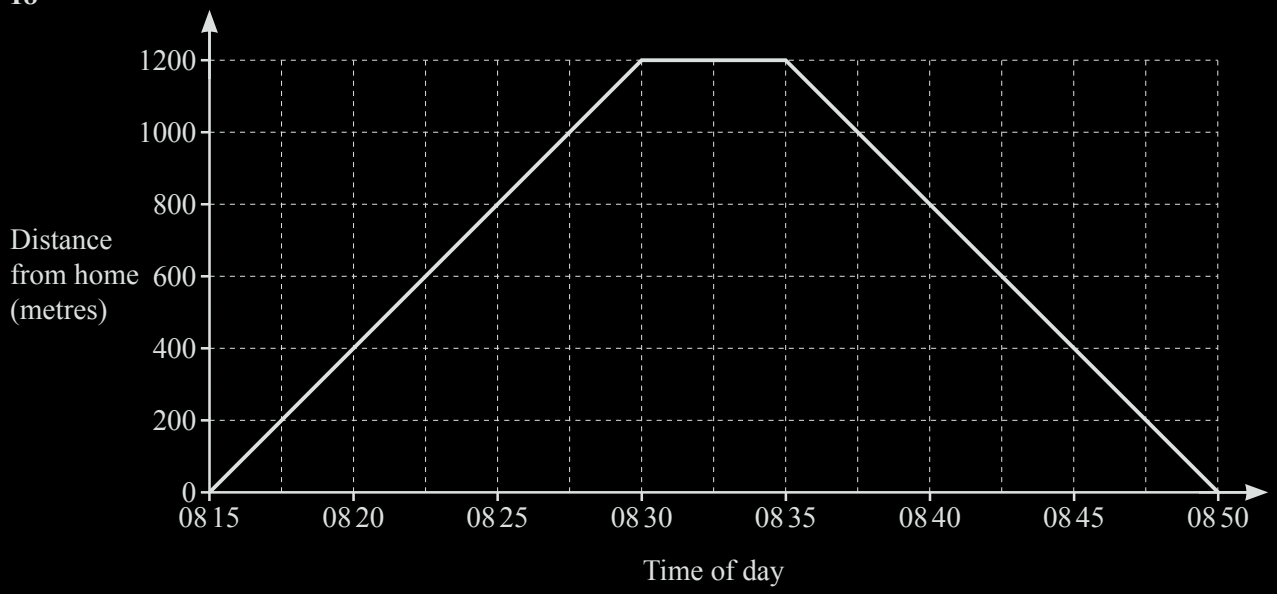
- nearer to $C$ than to $B$.

Shade the region in which this ship is situated. [1]

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**18**




The diagram is the distance–time graph of Safira's journey from home to a shop and back again.
She leaves home at 08 15 and returns at 08 50.

**(a)** How many minutes does she stay in the shop?

..... minutes [1]

**(b)** At 08 30, her brother leaves home and goes to the shop.
He walks at the same speed as Safira.

**(i)** On the grid, draw the graph of his journey to the shop. [1]

**(ii)** How far is he from the **shop** when he meets Safira?

..... m [1]

**(c)** Calculate the speed Safira walks to the shop.
Give your answer in km/h.

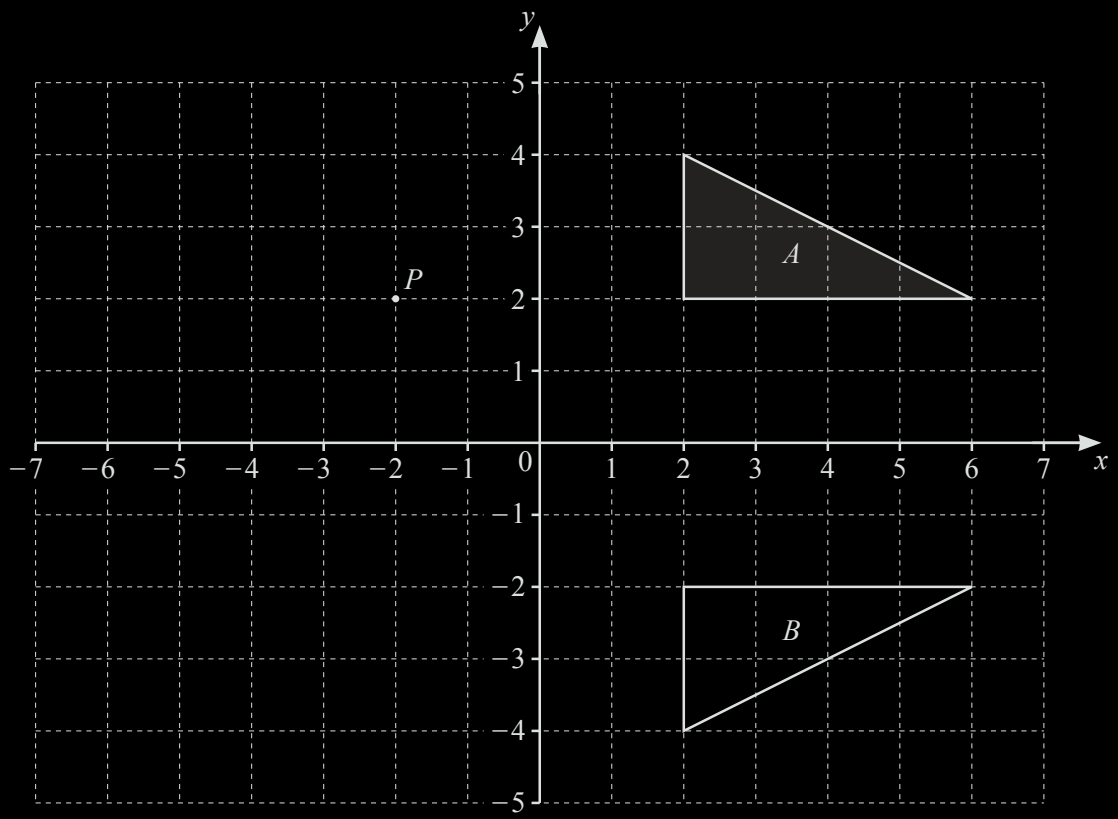
..... km/h [2]

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**[Turn over**

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**19**




Triangle $A$, triangle $B$ and the point $P$ (–2, 2) are drawn on the grid.

**(a)** **(i)** Describe, fully, the **single** transformation that maps triangle $A$ onto triangle $B$.

..... [2]

**(ii)** Write down the matrix that represents this transformation.

$\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

**(b)** Triangle $A$ is mapped onto triangle $C$ by an enlargement, centre $P$, scale factor $-\frac{1}{2}$.

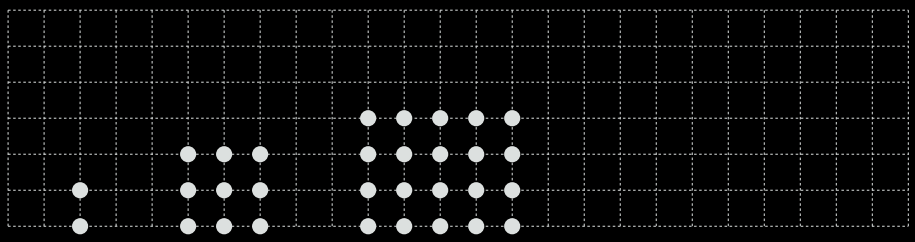
On the grid, draw and label triangle $C$. [2]

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**20**



Pattern 1 Pattern 2 Pattern 3 Pattern 4

The diagram shows a sequence of patterns.

Each pattern has one more row, and two more dots in each row, than the pattern before it.

**(a)** On the diagram, draw Pattern 4. [1]

**(b)** **(i)** Complete the table for the first four patterns in this sequence.

| Pattern number | 1 | 2 | 3 | 4 | | $n$ |
|---|---|---|---|---|---|---|
| Number of rows | 2 | 3 | 4 | | | $p$ |
| Number of dots **in each row** | 1 | 3 | | | | $q$ |
| **Total** number of dots | 2 | 9 | | | | |

[1]

**(ii)** Find an expression, in terms of $n$, for $p$.

$p =$ ..... [1]

**(iii)** Find an expression, in terms of $n$, for $q$.

$q =$ ..... [1]

**(iv)** Find an expression, in terms of $n$, for the **total** number of dots in Pattern $n$.

..... [1]

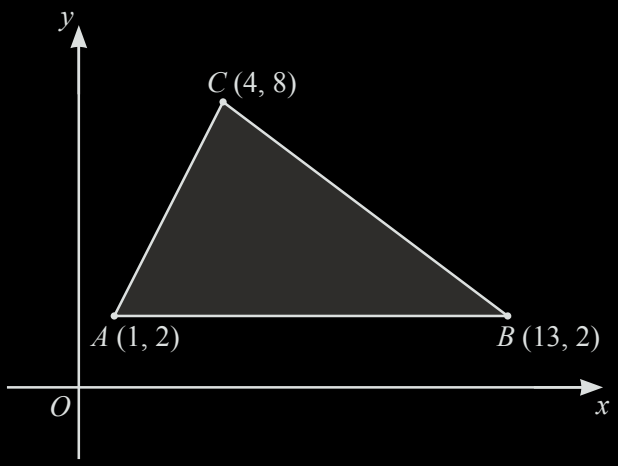
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**[Turn over**

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**21**




NOT TO SCALE

The diagram shows a triangle formed by joining the points $A$ (1, 2), $B$ (13, 2) and $C$ (4, 8).
The equation of the line $BC$ is $2x + 3y = 32$.

**(a)** The region **inside** triangle $ABC$ is defined by three inequalities.
One of these is $2x + 3y < 32$.

Write down the other two inequalities.

.....

..... [2]

**(b)** The point $(k, 7)$, where $k$ is an integer, lies **inside** triangle $ABC$.

Find the possible values of $k$.

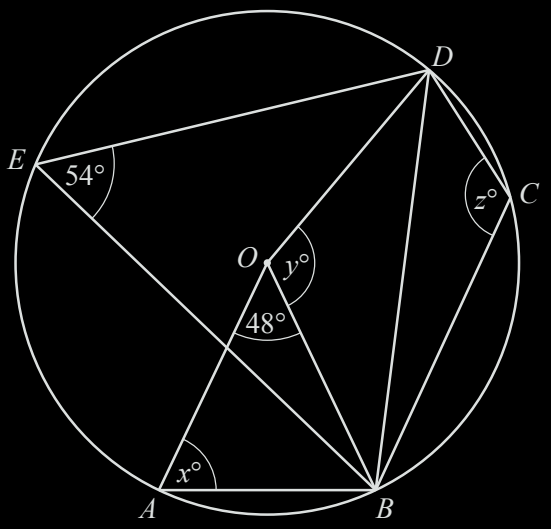
$k =$ ..... [2]

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**22**




NOT TO SCALE

In the diagram, $A$, $B$, $C$, $D$ and $E$ lie on the circle, centre $O$.
$A\hat{O}B = 48°$, $D\hat{E}B = 54°$.

**(a)** Find $x$.

$x =$ ..... [2]

**(b)** Find $y$.

$y =$ ..... [1]

**(c)** Find $z$.

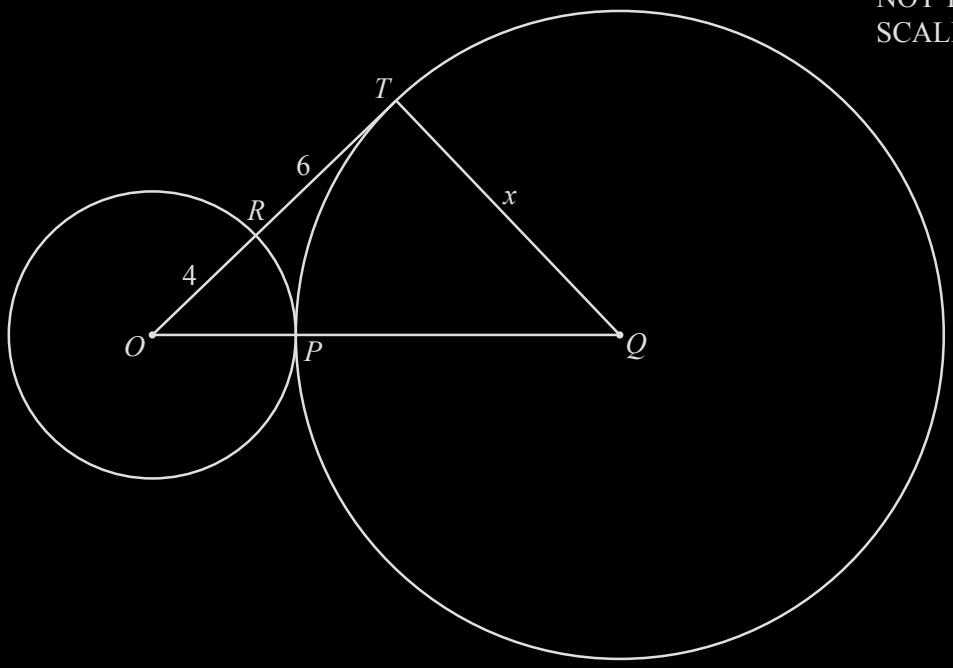
$z =$ ..... [1]



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**23**

NOT TO SCALE




In the diagram, the circles with centres $O$ and $Q$ touch at $P$ where $OPQ$ is a straight line.
The line $ORT$ intersects the smaller circle at $R$ and is a tangent to the larger circle at $T$.

$OR = 4$ cm and $RT = 6$ cm.
The radius of the larger circle is $x$ cm.

Calculate the value of $x$.

$x =$ ..... [4]

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**24** $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ -3 & -2 \end{pmatrix}$

**(a)** Find $\mathbf{A}^2$.

$\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

**(b)** The matrix $\mathbf{X}$ satisfies the equation $\mathbf{X}\begin{pmatrix} 2 & 1 \\ -3 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 2 \end{pmatrix}$.

Find $\mathbf{X}$.

$\mathbf{X} =$ [2]

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